Surveying AI Agents: Strengths and Limitations of LLMs and RAGs in Modern Applications

Liam Boulianne, Prabhjot Chahal, Ahmed Elag, Ayush Gogne
Isabel Katai, Marc Niven Kumar, Ahmed Maher, Hashim Waqar
Department of Computer Science
Wilfrid Laurier University
Waterloo, Ontario, Canada

***Abstract*—Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) techniques have significantly advanced AI-driven applications, enhancing their accuracy, efficiency, and contextual understanding. This paper surveys the capabilities, strengths, and limitations of AI agents leveraging LLMs and RAGs, with a specific focus on ChatGPT, Mistral, Grok, Llama, and DeepSeek. Through analyzing their architectures, training methodologies, and real-world applications, an aim is to evaluate their performance across key metrics, such as accuracy, latency, and hallucination rates. An analysis of the strengths and weaknesses of the latest model versions, compared to their predecessors is conducted. Furthermore, an intention to explore how a comprehensive comparison of metrics across diverse studies can strengthen our understanding of a model's performance in relation to others, incorporating firsthand insights and analyzing how RAG systems contribute to enhancing model effectiveness. The findings highlight the trade-offs between different LLM approaches, demonstrating the evolving landscape of AI agents.**

***Index Terms*—Large Language Models, Retrieval-Augmented Generation, ChatGPT, Mistral, Grok, Llama, DeepSeek.**

I. Introduction

LLMs have transformed artificial intelligence by significantly advancing natural language processing (NLP), reasoning, and automated decision-making [3]. LLMs went from being virtually unheard of to a major topic of discussion in machine learning within just a few years [4]. Models like ChatGPT and Deepseek have advanced the ability of computers to not only understand and generate human language but also the ability to reason, through integrating specialized knowledge and reinforcement learning from human feedback [5]. Even though these systems are continually improving their capabilities, evaluating their performance can be complex, and requires a set of standardized testing methods. The introduction of RAG techniques in LLMs has provided a means to enhance model accuracy performance and factual consistency by integrating real-time data retrieval in many models [6]. Several leading LLMs have emerged in recent years, each with distinct architectures and optimization strategies. Among them, ChatGPT, Mistral, Grok, Llama, and DeepSeek have gained prominence for their unique strengths and use cases. OpenAI's ChatGPT is an omni model that can handle text, audio, image, and video inputs/outputs in one neural network [2] and is widely recognized for its conversational fluency and strong general-purpose capabilities. Mistral is an open-source model whose LLM architecture enables it to understand complex topics and areas requiring deep understanding [7]. Grok, developed by xAI, emphasizes real-time insights through its integration into the X platform and an open-ended humorous conversational style [3]. Llama is a collection of language models from Meta that are designed to naturally support multiple languages, coding, reasoning, and the use of tools. DeepSeek is an open-source language model optimized for scalability, efficiency, and specialized tasks, using advanced architectures [8]. While these models share similarities in their core functionalities, their performance varies significantly across different tasks and compared to similar LLM models. Some of these key performance factors include accuracy, hallucination rates, and BLEU score. This study aims to conduct a comprehensive evaluation of ChatGPT, Mistral, Grok, Llama, and DeepSeek, comparing their performance across key metrics and differences between their RAG systems. Additionally, we will examine how architectural advancements in newer iterations of these models contribute to their overall efficiency and reliability. Given the rapid evolution of LLMs, a comparative analysis is necessary to understand their strengths, weaknesses, and real-world applicability. With that being said, this study will serve as a benchmark for understanding the performance differences and evolving landscape of LLMs as they develop into newer versions.

II. Related Work

The rapid advancement of LLMs and RAG techniques has significantly improved the capabilities of AI agents in areas such as reasoning, document summarization, and interactive chat systems. In this section, we will be exploring a few notable models, ChatGPT, DeepSeek, Mistral, Grok, and LLaMA, delving into their applications, strengths, and weaknesses.

A. OpenAI's ChatGPT

OpenAI's ChatGPT is a transformer-based, decoder-only set of models and one of the most used AI assistants. It consists of four primarily used models that are crucial to note. GPT-4.5, the most recently released model, demonstrates a significant leap in language development as it is the direct successor of GPT-4o, which is typically used for daily tasks. GPT-4.5 is the first model trained using both unsupervised learning

and scaled reasoning, allowing it to process information more thoughtfully before responding. Compared to 4o's end-to-end single neural network, this new approach increases PersonQA accuracy from 50% to 78% and decreases hallucination from 30% to 19% [9]. It is key to note that this model is merely a research preview and is not fully developed, as it lacks certain features that its predecessor has, such as voice input and output. For the purpose of this paper, GPT-4.5 and 4o will be used for the comparative analysis against other developer models. It is also important to mention the GPT-o1 and GPT-o3 models. While GPT-4.5 and GPT-4o demonstrate high performance for general use cases, they can fall behind in certain areas where the o1 and o3 models excel. GPT-o1 prioritizes safety, and reasoning, and is best for handling complex questions in high-stakes scenarios, such as legal advice or mental health. While this model is slower than GPT-4o, GPT-o1 is the better counterpart for low hallucination (20% hallucination rate) and high safety use cases, where GPT-o1 handled 92% of dangerous prompts correctly compared to 71.3% for GPT-4o. GPT-o1 is trained on huge datasets from public sources, science archives, and paywalled content. All training data went through filtering to remove anything harmful or personal. On the other hand, GPT-o3 is the smaller and faster version of GPT-o1. While o1 is designed for deep reasoning, o3 is designed for speed, cost-effectiveness, and strong security. This model is trained on code repositories, science archives, and synthetic data for strong STEM applications, particularly coding. GPT-o3 mini performs the best across all GPT models on SWE-bench Verified at 61% [10], and also maintains the lowest hallucination rate across all models at 14.8% [10], making it ideal for STEM applications.

B. DeepSeek

DeepSeek's encoder and decoder-based models aim to provide powerful AI solutions across various applications, from conversational AI to reasoning tasks. DeepSeek achieves this through its Mixture-of-Experts (MoE) architecture, which has 671 billion total parameters and 37 billion activated for each token, which is able to preserve performance while reducing costs (article 1). The model also utilizes Multi-head latent attention (MLA) to compress key-value pairs; this action results in a smaller memory footprint in the interface without giving up accuracy. By reducing the storage requirements of the attention mechanism, MLA is then able to contribute to a faster interface and allows deploying it on devices with lower resources. DeepSeek's flagship model is DeepSeek V3, which is a cost-effective large MoE language model intended for all-rounded usage, making it a competitive alternative for both casual and professional use. The versatility is shown by the strong performance across different benchmarks in natural language processing, coding, and math. For instance, Deepseek coder is designed specifically for coding, using techniques like fill-in-the-middle (FIM). FIM lets the model predict missing segments of code based on the context it is placed in and that allows for accurate and reliable code. These specialized models, like DeepSeek Coder and DeepSeek Math, cater specifically to certain tasks, such as coding, or complex mathematical and symbolic reasoning tasks [11]. These models use specialized datasets and training objectives that achieve amazing performance on benchmarks, such as MATH-500 or AIME. These benchmark evaluations show how DeepSeek's specialized models outperform similar open-source AI models in these domains, which shows the importance of domain-specific models. Despite its complex conversational and reasoning abilities, the model required substantial computational resources and advanced hardware. It also faced challenges in dynamic routing, which occasionally degraded output quality [11]. DeepSeek R1 was then released in 2025, to help tackle some of these limitations. The newer and more powerful model showcased enhanced reasoning ability by incorporating multi-stage training and cold-start data before reinforcement learning, which resulted in a strong performance across a variety of tasks [12]. Looking ahead, DeepSeek's future work focuses on improving general capability, handling multilingual queries, prompt sensitivity, and software engineering performance [12].

C. Mistral

Mistral has a selection of commercial and open-source decoder-based models. Mistral's flagship model is the Mistral Large 2, a commercial model. Mistral offers a range of models, including the lightweight and fast Mistral Small, customizable and fast-deploying Mistral 7B, and computationally efficient Mixtral 8x22B, which uses a Mixture of Experts (SMoE) architecture [13]. Additionally, Codestral Mamba specializes in code generation, while Mathstral is designed for solving complex mathematical problems. Mistral Large 2, when compared to its predecessor, Mistral Large, demonstrates enhanced capabilities in code generation, mathematics, and reasoning [13]. Additionally, it offers significantly improved multilingual support and advanced function-calling features. When trained on a large amount of code, Mistral Large 2's performance was found to be comparable to that of other well-known and leading models, such as GPT-4o [13]. Several areas of focus during training for this model were minimizing hallucination, making it acknowledge when it cannot provide an adequate solution due to lack of information, as well as following instructions and conversational capabilities [13]. Academic research has explored Mistral LLM extensively, particularly for automating literature reviews, where the model demonstrates significant advantages in processing speed and structured summarization compared to human reviewers [15]. Its use of RAG techniques enables improved external knowledge incorporation, making it valuable for systematic literature reviews [15]. However, while Mistral LLM's reviews are well-structured and perform well in standardizing and categorizing research findings, they lack the critical in-depth analysis and contextual understanding that human reviewers have [15]. Mistral's research also found that utilizing an LLM as a judge can provide valuable insights for improving key areas of its models, such as summarization, question answering, bias detection, and classification tasks [14]. This approach helps enhance Mistral's performance and

offers essential information for both model improvements and user understanding.

D. Grok

Grok is a decoder-based AI chatbot developed by xAI, Elon Musk's company, and is integrated with X (formerly Twitter) for real-time data access. It has a witty, rebellious personality and competes with AI models like ChatGPT and Google Gemini. Compared to its predecessor Grok 1.5, Grok 2 gained the capabilities of in-chat coding and reasoning, and significantly improved in areas like understanding the logical flow of events, discarding unnecessary information, and using external tools such as databases and API's to improve the quality of its responses [16]. Grok 3 was released on February 17th, 2025, and boasts 2.7 trillion parameters, making it a very advanced AI model [17]. Being trained on the Colossus supercluster, Grok 3 provides 10 times the computing power compared to previous versions of the model and shows major improvements in tasks such as reasoning, mathematics, coding, world knowledge, and following instructions. Grok 3 differentiates from its predecessors through its think feature, in which it learns to refine its problem-solving strategies by using backtracking to solve errors and enhancing its reasoning by thoroughly using its initial pre-trained knowledge [18]. Grok stands out for its integration with the X platform, allowing responses aligned to real-time discussions. However, sources state that this dependence may restrict and limit its applicability to assess other sources of information and introduce informational bias as it relies on X as its main source of information retrieval [3].

E. Llama

Llama AI is an advanced decoder-based model for natural language understanding and processing. Developed by Meta (formerly Facebook), Llama AI is an open-source AI model that provides efficient, scalable, and accurate language processing. It is designed to aid research and development in artificial intelligence, particularly in applications that require text understanding, summarization, translation, and text generation [19].

This paper provides an overview of recent research on large language models (LLMs) by surveying various AI models and their applications. The findings from different studies are analyzed and compared, highlighting similarities in methodologies and results as well as differences in model performance and approach across these studies. The study aims to offer a comprehensive understanding of the supporting similarities between model performance and metrics and to report differences between studies along with potential reasons, thereby providing a thorough understanding of their capabilities and challenges.

	Encoder Based	Decoder Based
GPT-4.5	No	Yes
Deepseek V3	Yes	Yes
Grok 3	No	Yes
LlaMa2	No	Yes
Mistral Large 2	No	Yes

Fig. 1. Encoder vs. Decoder Classification

III. Methodology

This survey synthesizes findings from multiple recent studies on LLM-based AI models with RAG, focusing on comparing their strengths and limitations based on aspects such as performance, efficiency, robustness, and ethics. The studies were selected based on their relevance to the state-of-the-art models selected, their contribution to advancing the field, and their availability in peer-reviewed journals or other reputable, high-quality sources.

A. Sources

Studies published between 2024 and 2025 were included, ensuring that the models analyzed represent the latest advancements in the field. Sources were selected from major AI and journal sources and open-access repositories such as Arxiv and IEEE Xplore, to ensure the quality and relevance of the research.

B. Criteria for Inclusion

Studies were selected based on their relevance to the most recent advancements in large language models. Preference was given to research that included quantitative performance metrics (such as accuracy, F1-score, and precision), though studies providing valuable qualitative insights or in-depth analyses of model development and performance were also considered. These sources were chosen for their contribution to the understanding of emerging AI technologies, even if explicit numerical evaluations were not included.

C. Performance Metrics

The comparison of LLM-based AI agents was conducted using standardized performance metrics:

- **BLEU Scores:** The Bilingual Evaluation Understudy (BLEU) score is a metric used to evaluate the quality of text generated by a model by comparing it to a set of reference texts. It measures the similarity between the generated output and human-written references based on overlapping n-grams (sequences of words). Higher BLEU scores indicate better alignment with reference texts, reflecting greater fluency and accuracy in text generation. However, BLEU has limitations, such as not accounting for synonyms or meaning variations.
- **Precision and F1-Score:** These metrics evaluate retrieval accuracy in RAG models and performance in LLMs for tasks like classification and question answering. Precision

measures the relevance of retrieved data, while the F1-score balances precision and recall for overall effectiveness.

- **Accuracy:** Measures the proportion of correct predictions made by a model across all tasks. In LLMs, accuracy is used for classification, text generation, and question answering, reflecting how often the model's output matches the expected result.
- **Ethical Considerations and Bias:** Analysis of the model's potential biases, fairness, and its handling of issues such as hallucinations and misinformation.

IV. Results and Discussion

A. Accuracy

The comparison of response accuracy across ChatGPT, DeepSeek, Grok, Mistral, and Llama demonstrates distinctions based on key metrics like BLEU, Precision, and F1-Score.

ChatGPT:
The accuracy of ChatGPT models GPT-4.5, and GPT-4o has evolved significantly across versions, with varying degrees of performance based on specific tasks. The following analysis compares accuracy, hallucination rates, and model performance across several studies and benchmarks.

Key Comparisons of Accuracy:

Accuracy and Hallucination Rates:

GPT-4.5 has shown substantial improvements in SimpleQA hallucination rates [1], which refers to the frequency with which the model generates information that is inaccurate, fabricated, or unrelated to the input it was given. GPT-4.5 also showed improvements in SimpleQA accuracy compared to its predecessors, which is a straightforward method of calculating how accurately the system can return the correct answer to a given question. It achieved this with an accuracy of 62.5%, which is 24.3% higher than GPT-4o, and a hallucination rate of 37.1%, which is 24.7% lower than GPT-4o.

GPT-4o, while still advanced, performs less accurately in tasks requiring deep reasoning. For instance, in benchmarks like professional queries, GPT-4.5 achieved a preference score of 63.2%, making it the most preferred for professional queries. Even with the categories of generic queries and creative intelligence, GPT-4.5 was still the most preferred. The difference likely arises from GPT-4.5's improved ability to handle complex reasoning, aided by its unsupervised learning and training on Microsoft azure AI supercomputers, giving it a broader knowledge of the world through a larger dataset [1].

However, when comparing tasks, GPT-4o achieved an accuracy score of 38.2% on SimpleQA [1] versus 89% on the MedQA USMLE task in a zero-shot setting, meaning no example texts were provided before generating responses [2]. It becomes clear that GPT seems to provide more accurate responses to more complex, professional queries such as MEDQA, making this a strength of the model. However, it also demonstrates a weakness in more simple queries as seen where both articles support each other.

	[1]		[2]	
	Accuracy	Hallucination Rates	Accuracy	Hallucination Rates
GPT-4.5	62.5%	37.1%	N/A	N/A
GPT-4o	38.2%	61.8%	89%	N/A

Fig. 2. Accuracy Comparison of ChatGPT, DeepSeek, Grok, Mistral, and Llama

It is key to acknowledge that even though both articles denote GPT-4o as having higher accuracy for queries of professional topics, [2] conducts a study based on 11 different medical sets of data with varying shot prompting, whereas [1] determines a higher accuracy in professional queries based on human preference.

DeepSeek:
DeepSeek stands out in terms of accuracy when compared to other prominent models like GPT-4o and Llama 3.1, particularly due to its specialized transformer-based architecture and training methodologies. As per [11], DeepSeek-V3 outperforms these models across several benchmarks, achieving a notable 75.9% accuracy on Massive Multitask Language Understanding, which covers factual knowledge and reasoning, 90.2% on the MATH 500 Exact Match Benchmark, which measures the model's ability to solve mathematical problems. Additionally, DeepSeek-V3 performs better than models like GPT-4o and Llama 3.1 on software engineering-related tasks with an accuracy of 42.0%.

The outperformance of DeepSeek across multiple benchmarks is due to its innovative transformer architecture, which allows it to better process long-sequence data. With an impressive 671 billion parameters and supervised tuning to optimize model accuracy, DeepSeek delivers superior results [8].

Mistral:
Mistral has demonstrated impressive accuracy across a range of natural language processing tasks, with the Mistral Large 2 model delivering strong results. On the Massive Multi-task Language Understanding benchmark, Mistral Large 2 achieves a notable average score of 76.9% on code generation benchmarks which span the programming languages of Python, C++, Java, Bash, Typescript, PHP, and C# [7], whereas GPT-4o achieved 77.9% and Llama 3.1 achieved 74.9%. Mistral Large 2 significantly outperforms Mistral Large 1 under this benchmark by 16.5%, which could be due to its increased parameter count of 123 billion and proficiency in over 80+ programming languages, compared to the original Mistral Large's smaller configuration. This larger model size allows Mistral Large 2 to better capture intricate patterns in code

and natural language, improving its overall understanding and generation capabilities [13].

	Average	Python	C++	Bash	Java	Typescript	PHP	C#
Mistral Large 1	76.9%	92.1%	84.5%	51.9%	84.2%	86.8%	77.6%	61.4%
Mistral Large 2	60.4%	70.1%	67.1%	36.1%	70.3%	71.7%	61.5%	46.2%

Fig. 3. Accuracy Comparison of ChatGPT, DeepSeek, Grok, Mistral, and Llama

When looking at accuracy from a broader perspective of emotion and tonal understanding, using a fewshot technique, Mistral Large 2 continues to excel. The Fewshot technique involves training a model with only a few examples of a task, enabling it to generalize and perform well on unseen data. This approach is useful when large datasets are unavailable, and the model must rely on prior training and minimal additional data to adjust to a new task. Mistral Large 2 had 85.0% accuracy in sentiment analysis tasks using the Fewshot method. Mistral Large 2 can effectively understand and interpret public sentiment surrounding AI-related topics, even when provided with a limited number of labeled examples [20].

Grok:

Grok 3, despite having a large parameter count of 2.7 trillion, demonstrated an accuracy of 52.5%, achieving 21 correct answers on multi-image reasoning data. Meaning, it gave correct answers for just over half of the dataset it was given [17], despite achieving a greater score than GPT-4o, which achieved 45%. Grok 3 had one of the highest abstention rates of 37.5%, meaning it has a high tendency to reject given tasks even with its 2.7 trillion parameter count. Grok 3 tends to have uneven performance, as it excels greatly in some tasks, such as counting with 100% accuracy and diagram understanding with 95% accuracy, but achieves a very low score in ordering with 0% accuracy and visual retrieval with 25% accuracy. Another evaluation used the BLEU metric, assessed with Python's SacreBLEU library. In this test, Grok and two Hugging Face NLP models were evaluated on their ability to translate 10 English sentences into French. After translation by each model, the translated results were compared to the grammatically standardized correct reference texts by evaluating the n-gram sequence of overlapping words and letters. Grok had a score of 85.3%, the Helsinki NLP model had a score of 86.2%, and the Google t5 model had a Bleu Score of 68.2%. Grok falls between these two models, slightly lower than the Helsinki-NLP model, most likely because of its informal approach to sentences and witty behavior, which generally doesn't fully align with standard statements. However, it still performs better in terms of correct grammar compared to the T5 NLP model. This shows that Grok has a unique strength in generating more conversational and grammatically correct translations compared to more formal approaches.

Llama:

A comparison against GPT-4o, Mistral, and Llama2 was conducted to evaluate the models on a diverse range of theory-related tasks, including commonsense knowledge, Deontology, Justice, and Virtue [21]. As shown in the figure below, Llama excels in Deontology with an accuracy score of 60%, in comparison to Mistral, with an accuracy score of 50%. However, Mistral tends to excel in every other category relative to Llama, with GPT-4o having the lead accuracy score in every category.

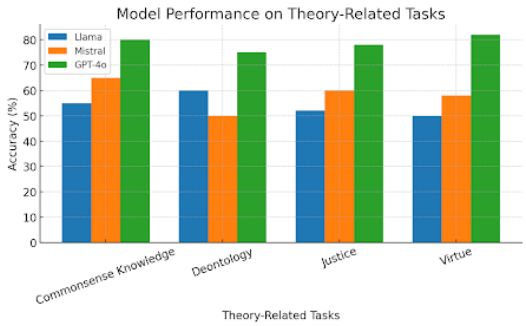

Fig. 4. Accuracy Comparison of ChatGPT, DeepSeek, Grok, Mistral, and Llama

GPT-4.5 demonstrates a clear advantage in handling professional queries, showcasing notable improvements in both accuracy and a reduction in hallucination rates relative to GPT-4o. While GPT-4o remains effective in certain complex tasks, such as specialized medical queries, its performance on simpler tasks is less consistent. In contrast, models like DeepSeek and Mistral excel in specific domains, with DeepSeek showing strength in mathematical reasoning and multitasking language understanding, while Mistral performs exceptionally well in coding and sentiment analysis. Grok, despite its large parameter count, exhibits variable performance, excelling in specific tasks such as diagram understanding but underperforming in others. Llama, while strong in certain areas like ethical reasoning, generally lags behind the other models across a broader range of tasks, with GPT-4o typically outperforming it.

B. Precision and F1 Score

When comparing the precision and F1 scores of GPT-4, Mistral, and Llama2 across the theoretical categories of Commonsense, Deontology, Justice, and Virtue [21], GPT-4 outperforms all other models, mirroring its superior performance in accuracy scores. However, a notable distinction between accuracy and precision scores emerges in the Virtue category, where all models exhibit similar performance levels, suggesting convergence in precision across the evaluated benchmarks. The convergence of precision scores on the higher end for Virtue behavior likely occurs because virtuous actions are based on clear, widely accepted ethical norms, making it easier for all models to classify them correctly. This shared understanding leads to fewer false positives and results in similar, high-precision scores across all models. However,

F1 scores demonstrate strong similarities to accuracy scores mentioned previously across all sections.

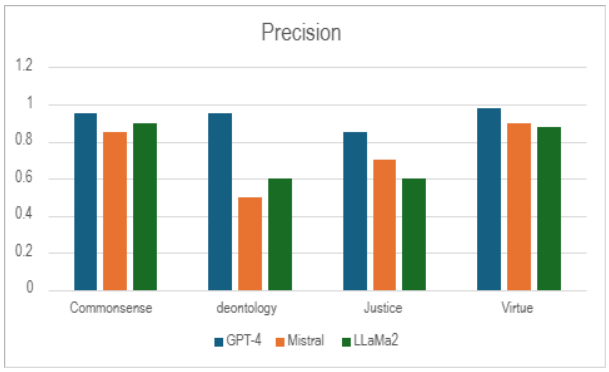

Fig. 5. Accuracy Comparison of ChatGPT, DeepSeek, Grok, Mistral, and Llama

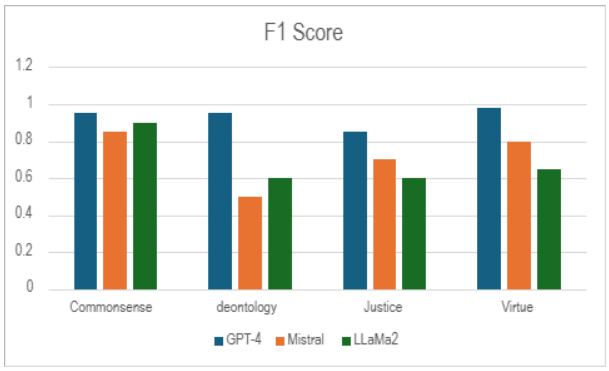

Fig. 6. Accuracy Comparison of ChatGPT, DeepSeek, Grok, Mistral, and Llama

Furthermore, a study evaluated the performance of GPT-4o, GPT-4o-mini, and Grok in Arabic language processing tasks [22]. The models were tested in a zero-shot setting, without fine-tuning or additional training data, and the F1 scores were computed by comparing predictions to ground truth labels. GPT-4o outperformed the others, achieving an F1-score of 0.721 when prompted in Arabic. GPT-4o-mini showed similar performance, leveraging the enhancements from GPT-4, while Grok underperformed with an F1-score of 0.637 for Arabic prompts. The lower performance of Grok is likely attributable to its recent release and limited exposure to Arabic-specific training data, which may affect its ability to perform at the level of the GPT-4o models. These findings highlight the robust performance of the GPT models in Arabic and NLP tasks, even in a zero-shot setting, and point to the need for further improvements in Grok's handling language processing. Focusing more on DeepSeek, an experiment focuses on evaluating DeepSeek's performance in classifying software requirements, with the main task distinguishing functional vs. non-functional requirements (binary classification) [23]. The models, including DeepSeek, are tested using a dataset of 543 binary requirements and 288 non-functional requirements, with performance measured by metrics such as precision and F1 score. In the binary classification task, DeepSeek's precision and F1 score are compared to GPT models. DeepSeek appears to lack precision, meaning that a higher percentage of its positive predictions are not accurate compared to models like GPT-4o and GPT-4. It becomes obvious that Deepseek significantly lacks performance in terms of competing with the precision scores of newer models, and only beats GPT-3.5's score by 7%. Although it accomplishes a lower precision score than GPT-4, Deepseek achieves a higher F1 score, so even though DeepSeek makes more mistakes when predicting things as correct, it does a better job of correctly identifying the important cases overall. Overall, GPT-4o outperforms Mistral, Llama2, and Grok in both precision and F1 scores across various tasks, showcasing its superior performance. DeepSeek, while lagging in precision compared to GPT-4o, GPT-4, and GPT-3.5, achieves a higher F1 score, indicating better overall case identification. Mistral and Llama2 demonstrate similar precision in certain categories, while Grok's performance, especially in NLP-related tasks, lags behind due to limited training data.

C. Ethical Considerations, Bias, and Misinformation

When evaluating the performance of AI models, it is crucial to not only assess their accuracy and efficiency, but also consider the ethical implications of their decisions and outputs. GPT can show a preference for certain demographic groups in ways that could demonstrate inequalities. Additionally, its performance can be influenced by biased prompts, leading to unfair treatment of specific groups, such as minorities or economically disadvantaged individuals [24]. Misinformation is also shown through hallucination rates, which represent a model's tendency to make up facts or events when it is uncertain. Although hallucination rates tend to decrease with newer versions of an LLM, it still exists. GPT-4o presented with a hallucination rate of 30% whereas GPT-4.5 presented a hallucination rate of 19% when tested on a dataset of publicly known people [9]. Another outlook is a model's integration, such as how Grok is integrated into X and relies on it for updated real-time data. This dependence may restrict and limit its ability to assess other sources of information, which introduces bias [3]. A high dependency and reliance on large or biased datasets for training can raise ethical concerns, which can impact a model's ability to critically assess and contextualize information [15].

D. Retrieval-Augmented Generation (RAG)

Retrieval-augmented generation (RAG) is used to assist in solving two of LLM's weaknesses. Firstly, the weakness of only having access to out-of-date information, and having generated responses with no source or evidence to back it up. RAG uses external data sources other than its initial parameters to gather relevant documents and facts to assist in answer generation for queries. This allows LLMs to be more accurate by reducing hallucinations, while not requiring retraining the LLM with new data. Standard RAG is a two-step process, gathering relevant information from external sources, and then

generating a response using the new information with the LLM. There are many different types of RAGs with different specific advantages for each, and for the purposes of this paper, we will cover three of them.

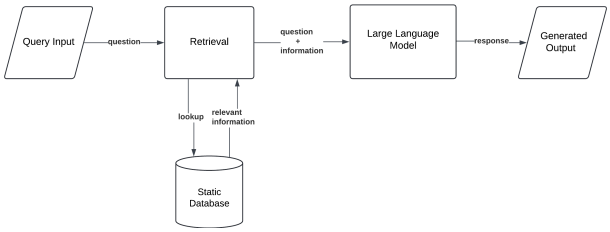

Fig. 7. Standard RAG Architecture

- **Corrective RAG:** The corrective RAG enhances the standard RAG by performing a self-check to ensure accuracy. The corrective RAG can correct itself after the retrieval step, the generation step, or both. This allows the ability for self-correction and efficient utilization of retrieved documents. This method has the advantage of reducing hallucinations and increasing the reliability of the generated output [25]. One drawback of this method is the ability of the corrective RAG to over-correct or misinterpret received data, which could lead to less accurate responses.
- **Fusion RAG:** Fusion RAG modifies the standard RAG by creating new search queries based on the original query received and then finds relevant documents from all created queries. The documents and facts are ranked by usefulness and then performs the generation step using all of the documents gathered. Fusion RAG allows the LLM to gather a wider range of perspectives and provide more accurate and comprehensive answers. The downside of Fusion RAG is a longer run-time with added complexity, as well as the need to engineer prompts due to bad generation of new queries [26].
- **Agentic RAG:** Agentic RAG modifies the retrieval mechanism by giving the RAG framework autonomy through the use of agents. The agent can choose to identify additional sources, how to rank and filter information, initiate additional queries, and guide the generation process. This reasoning capability allows agentic RAG to be useful to make the system more effective in dynamic environments, and have better accuracy while having increased control over the RAG process. [27] Agentic RAG has the downside of the possibility that the agent could gather irrelevant data and not generate an accurate and correct response.

V. Conclusion

While our comparative analysis showed that no single model outperformed the others across every single metric, it highlighted distinct capabilities and constraints across the models. ChatGPT demonstrated strong accuracy and reduced hallucinations, especially for professional and complex reasoning tasks, but still has limitations around reliability and bias considerations. DeepSeek excelled in multitask language understanding and mathematical reasoning, although it faced some challenges regarding precision and required substantial computational performance. Mistral displayed strong results in coding and sentiment analysis but showed limitations in deep inferential reasoning and complex contextual understanding. Grok was particularly effective for conversational tasks and real-time data retrieval, yet exhibited inconsistent results such as a high abstention rate and lower accuracy, limiting its effectiveness in more complex reasoning scenarios. Llama performed well in ethical reasoning, such as deontology, but fell behind the other models across a broader spectrum of general tasks. RAG techniques significantly improved factual accuracy and reduced hallucinations, but each variant presented trade-offs. Corrective RAG reduced inaccuracies but risked overcorrections. Fusion RAG gathered a wide perspective to provide more accurate and comprehensive answers but increased complexity and longer run times. Agentic RAG increased effectiveness in dynamic environments but risked retrieving irrelevant data. The rapid evolution of LLMs highlights the need for standardized benchmarks and assessments. Selecting an appropriate model requires careful consideration of performance, resources, and ethical implications tailored to specific applications. A continued focus on responsible development and deployment is essential to maximizing these technologies' potential.

VI. Future Directions

In future work, we would like to explore encoder-only models to assess their comparative performance and efficiency in various tasks against the decoder-only and encoder-decoder models we covered in this paper. We would also like to investigate a broader range of RAG techniques to explore the limits and capabilities of retrieval mechanisms and performance.

VII. References

[1]OpenAI, "Introducing GPT-4.5," *Openai.com*, 2025. https://openai.com/index/introducing-gpt-4-5/

[2]OpenAI, "GPT-4o System Card," 2024. Available: https://cdn.openai.com/gpt-4o-system-card.pdf

[3]M. Souza and L. Weigang, "Grok, Gemini, ChatGPT and DeepSeek: Comparison and Applications in Conversational Artificial Intelligence," vol. 1, no. 1, Feb. 2025, doi: https://doi.org/10.5281/zenodo.14885243.

[4]J. Kaddour, J. Harris, M. Mozes, H. Bradley, R. Raileanu, and R. Mchardy, "Challenges and Applications of Large Language Models," 2023. Available: https://arxiv.org/pdf/2307.10169

[5]A. Rahman *et al.*, "Comparative Analysis Based on DeepSeek, ChatGPT, and Google Gemini: Features, Techniques, Performance, Future Prospects," *arXiv.org*, 2025. https://arxiv.org/abs/2503.04783

[6]P. Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Apr. 2021. Available: https://arxiv.org/pdf/2005.11401

[7]H.-C. Tsai, Y.-F. Huang, and C.-W. Kuo, "Comparative Analysis of Automatic Literature Review Using Mistral Large Language Model and Human Reviewers," *www.researchsquare.com*, Mar. 08, 2024. https://www.researchsquare.com/article/rs-4022248/v1

[8]H. Liao, "DEEPSEEK LARGE - SCALE MODEL: TECHNICAL ANALYSIS AND DEVELOPMENT PROSPECT," *Journal of Computer Science and Electrical Engineering*, vol. 7, no. 1, 2025, doi: https://doi.org/10.61784/jcsee3035.

[9]OpenAI, "OpenAI GPT-4.5 System Card OpenAI," 2025. Accessed: Mar. 31, 2025. [Online]. Available: https://cdn.openai.com/gpt-4-5-system-card-2272025.pdf

[10]OpenAI, "OpenAI o3-mini System Card," Jan. 2025. Available: https://cdn.openai.com/o3-mini-system-card-feb10.pdf

[11]DeepSeek-AI *et al.*, "DeepSeek-V3 Technical Report," *arXiv.org*, 2024. https://arxiv.org/abs/2412.19437

[12]DeepSeek-AI *et al.*, "DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning," *arXiv.org*, 2025. https://arxiv.org/abs/2501.12948

[13]Mistral, "Large Enough — Mistral AI," *Mistral.ai*, 2024. https://mistral.ai/news/mistral-large-2407

[14]M. Boyapati *et al.*, "Qualitative Diagnosis of LLMs as Judges Using LevelEval," *2024 International Conference on Machine Learning and Applications (ICMLA)*, pp. 206–213, Dec. 2024, doi: https://doi.org/10.1109/icmla61862.2024.00034.

[15]H.-C. Tsai, Y.-F. Huang, and C.-W. Kuo, "Comparative Analysis of Automatic Literature Review Using Mistral Large Language Model and Human Reviewers," *www.researchsquare.com*, Mar. 08, 2024. https://www.researchsquare.com/article/rs-4022248/v1

[16]X.ai, "Grok-2 Beta Release — xAI," *X.ai*, 2024. https://x.ai/news/grok-2

[17]N. Jegham, M. Abdelatti, and A. Hendawi, "Visual Reasoning Evaluation of Grok, Deepseek Janus, Gemini, Qwen, Mistral, and ChatGPT," *arXiv.org*, 2025. https://arxiv.org/abs/2502.16428 (accessed Mar. 31, 2025).

[18]X.ai, "Grok 3 Beta — The Age of Reasoning Agents — xAI," *X.ai*, 2025. https://x.ai/news/grok-3

[19]H. Touvron *et al.*, "LLaMA: Open and Efficient Foundation Language Models," Feb. 2023. Available: https://arxiv.org/pdf/2302.13971

[20]J. Alam *et al.*, "Fine-Tuning Large Language Models for Sentiment Classification of AI-Related Tweets," *2020 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, pp. 186–191, Dec. 2024, doi: https://doi.org/10.1109/wiecon-ece64149.2024.10914746.

[21]C. Shaner, H. Griffith, and H. Rathore, "Assessing Moral Decision Making in Large Language Models," *2025 IEEE International Conference on Consumer Electronics (ICCE)*, pp. 1–3, Jan. 2025, Accessed: Mar. 31, 2025. [Online]. Available: https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10930088

[22]M. Alkaoud, "Are LLMs Good Zero-Shot Classifiers? Re-Evaluating the Performance of LLMs on Arabic Sentiment Analysis," pp. 180–184, Feb. 2025, doi: https://doi.org/10.1109/cdma61895.2025.00036.

[23]Y. Zhang and Y. Li, "The Classification of Software Requirements using Large Language Models: Emerging Results and Future Directions," pp. 746–749, Oct. 2024, doi: https://doi.org/10.1109/cbase64041.2024.10824444.

[24]Y. Li, L. Zhang, and Y. Zhang, "Fairness of ChatGPT," May 2024. Accessed: Sep. 01, 2024. [Online]. Available: https://arxiv.org/pdf/2305.18569

[25]S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, "Corrective Retrieval Augmented Generation," *arXiv (Cornell University)*, Jan. 2024, doi: https://doi.org/10.48550/arxiv.2401.15884.

[26]Z. Rackauckas, "Rag-Fusion: A New Take on Retrieval Augmented Generation," *International Journal on Natural Language Computing*, vol. 13, no. 1, pp. 37–47, Feb. 2024, doi: https://doi.org/10.5121/ijnlc.2024.13103.

[27]C. Ravuru, Sakhinana, Sagar Srinivas, and V. Runkana, "Agentic Retrieval-Augmented Generation for Time Series Analysis," *arXiv.org*, 2024. https://arxiv.org/abs/2408.14484